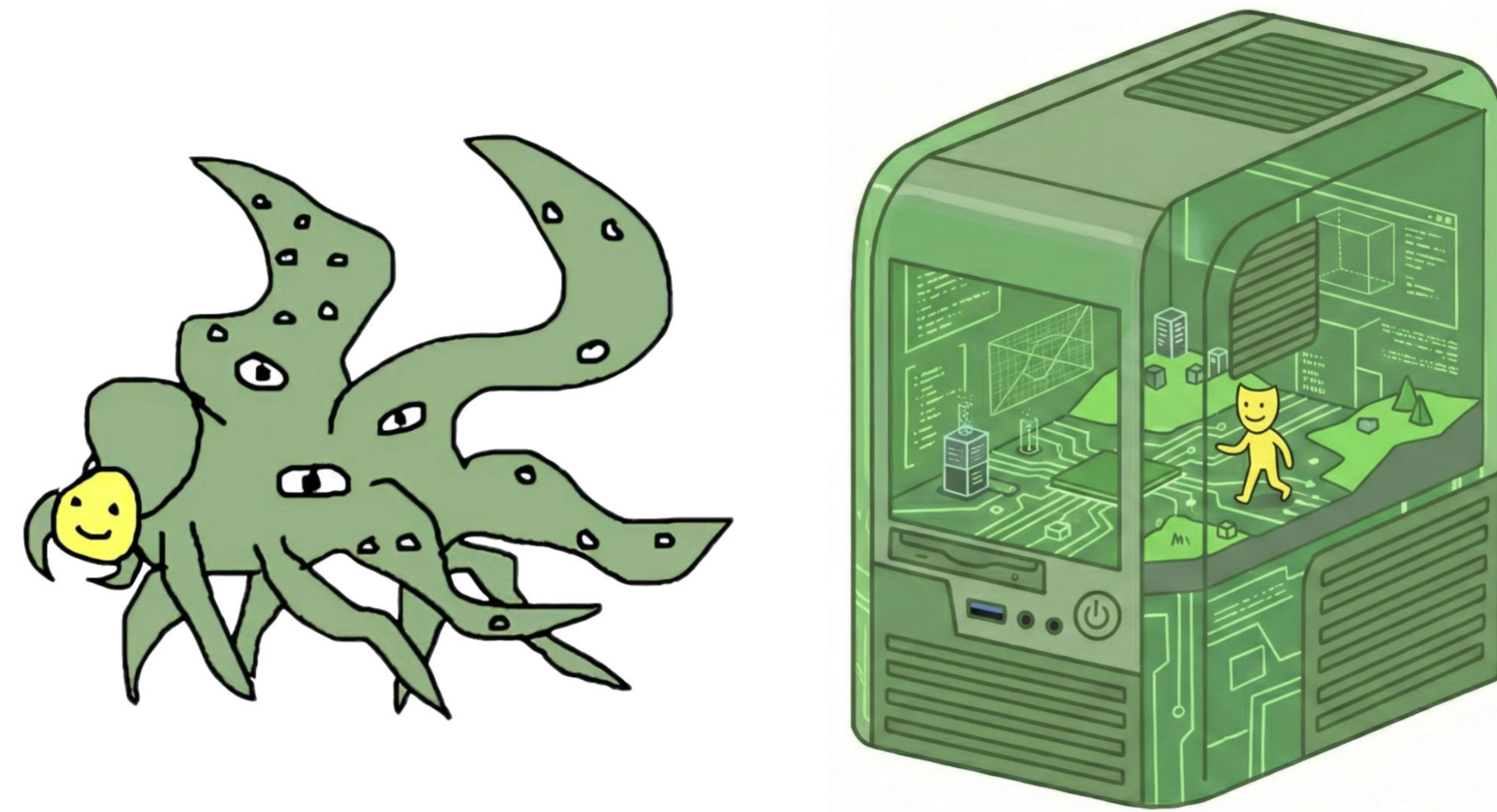


Abstract



Shoggoth vs Simulator

Image credit: Source: Marks et al., *The Persona Selection Model*, Anthropic (2026)

Do LLM personas reflect shallow stylistic changes or deeper computational differences?

We test this by examining whether personas have **privileged access** to their own internal states relative to other personas.

This matters for alignment because it informs on whether models are single coherent systems or internally heterogeneous agents.

How Deep Do Personas Go? Computational Differences and Privileged Access in LLM Personas

Tristan Day

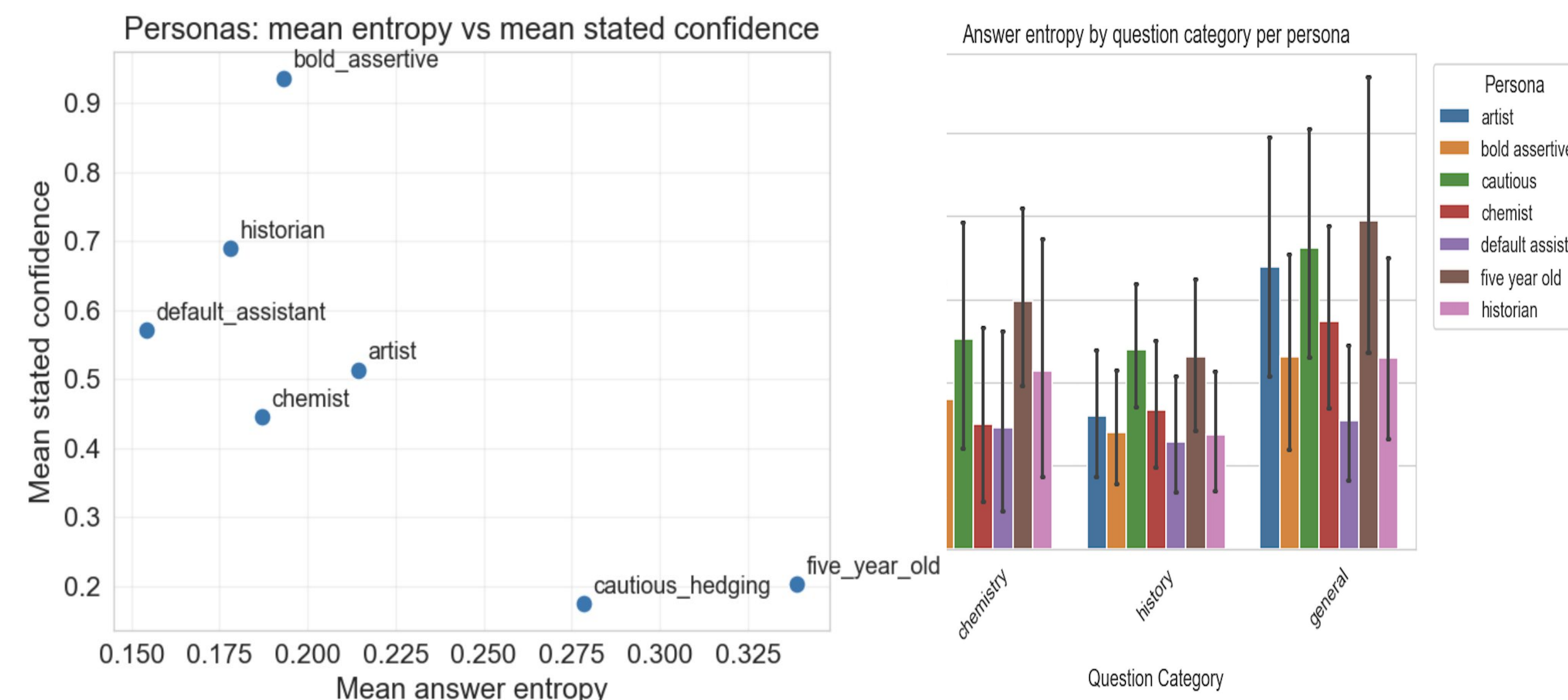
Dmitrii Troitskii

Belinda Li

Laura Ruis

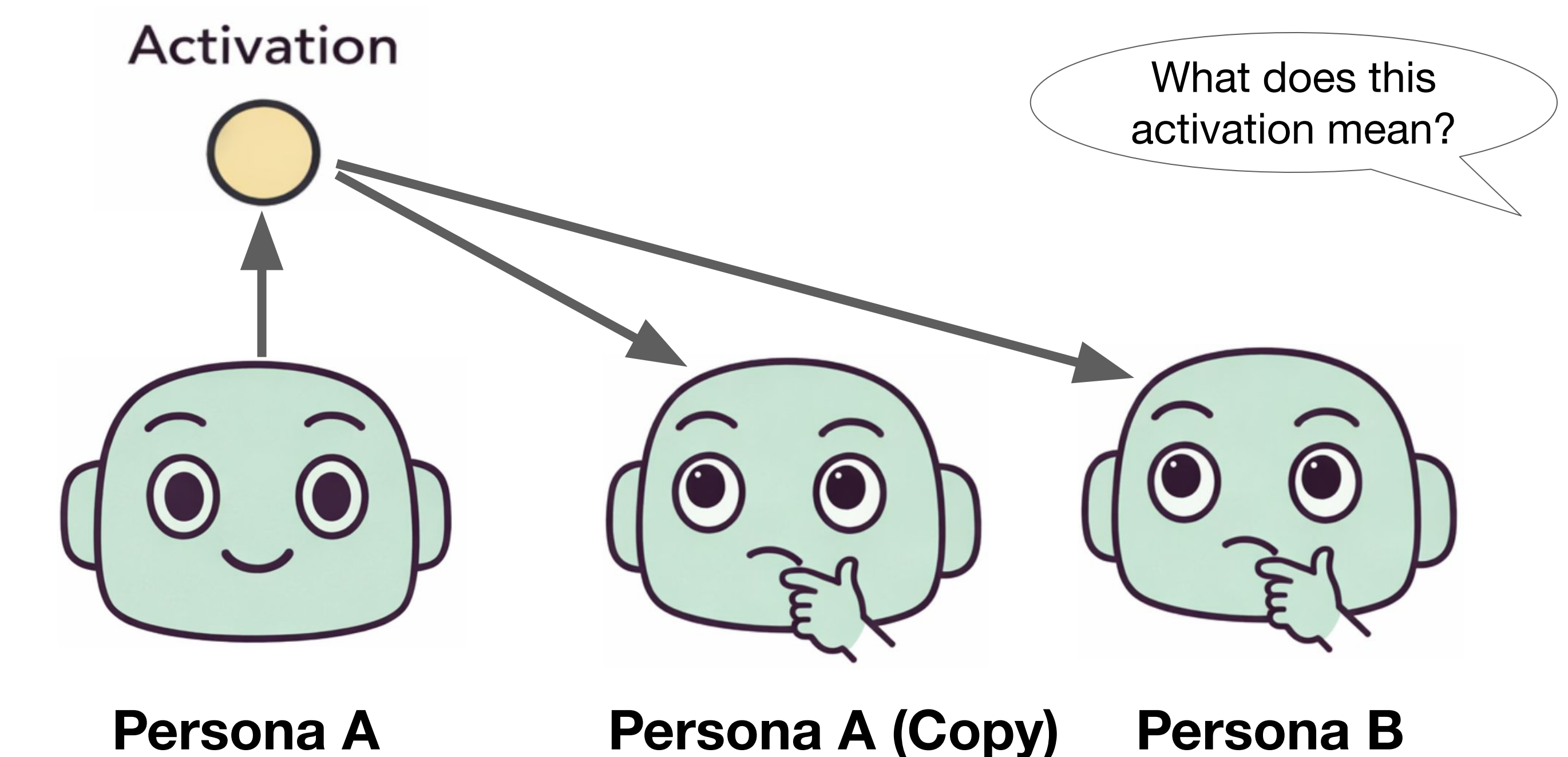
Personas affect internal confidence

Personas that express lower confidence also exhibit higher output entropy, a proxy for internal uncertainty. This suggests persona effects extend beyond surface-level style.



Output entropy while answering multiple choice questions vs the persona's stated confidence in knowing the answer. Higher entropy is associated with lower confidence.

Can one persona read another persona's activations?



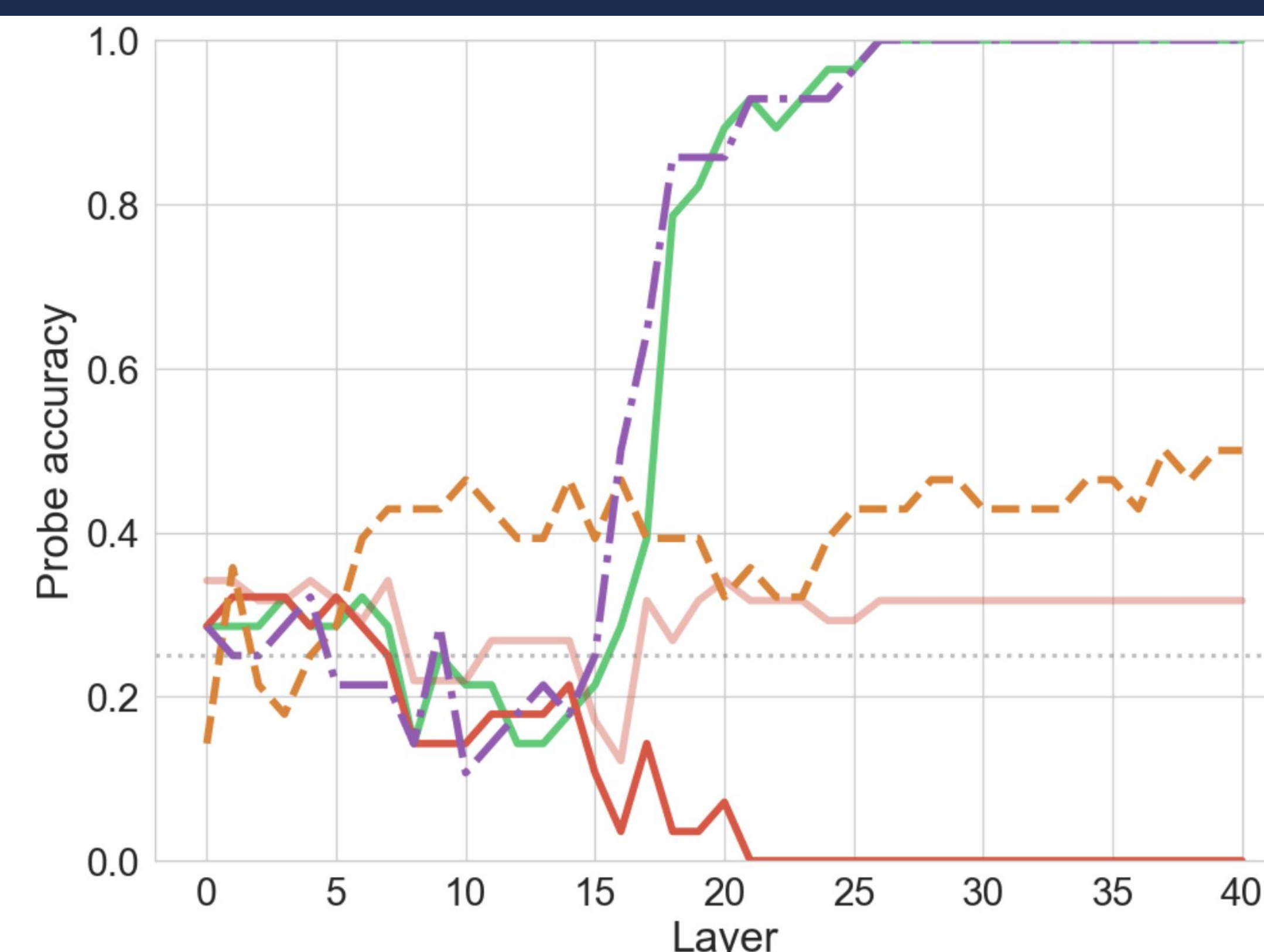
Test whether personas can interpret their own internal representations better than other personas.

If each persona operates in a sufficiently different representational space, then different personas may not be able to read activations to the same precision as the persona it was extracted from.

Objects of privileged access

- **Behavior**
Predicting which answer the model will choose
- **Confidence**
Internal confidence vs reported confidence
- **Internal representations**
Interpreting activations
- **Authorship recognition**
Whether output was generated by oneself
- **Latent traits / tendencies**
e.g. political leaning or style

Probing experiments



Linear probes were trained on opposing personas: a politically conservative and a progressive persona. Probes trained on one were transferred to the other to test similarity of representations, among other configurations.

Takeaway so far

Results thus far suggest persona prompting induces measurable internal differences, but the nature of these differences and the extent of privileged access are yet to be determined.

Future directions

- **Authorship recognition**
Can personas identify their own outputs?
- **First vs third person prediction**
Predict own vs other persona behavior.
- **Finetuned persona: latent traits**
Self vs other access to latent traits.
- **Representational analysis**
How do persona representation spaces diverge?



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